

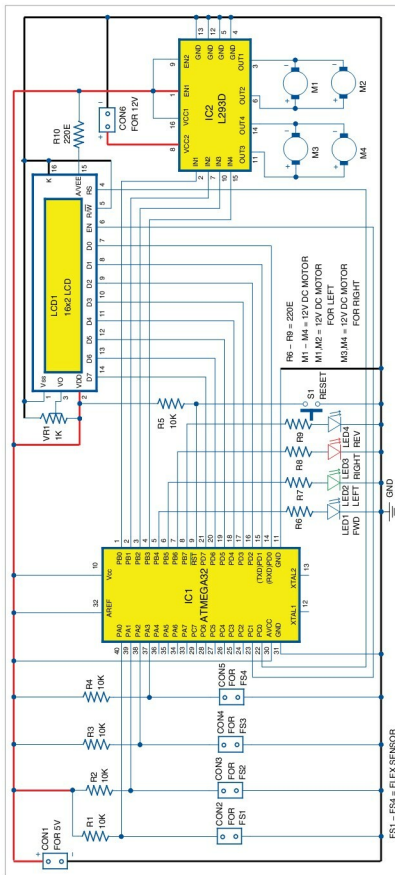


Fig. 4: Circuit diagram for gesture control of the wheelchair

configured in pull-up configuration with 10-kilo-ohm pull-up resistors. Flex sensor outputs are connected to ADC input pins 37 through 40 (PA3 to PA0) of IC1.

Port-D pins PD0 through PD7 of IC1 drive data pins D0 through D7 of LCD1. Control pins RS and EN of LCD1 are connected to port-C pins PC0 (pin 22) and PC1 (pin 23) of IC1, respectively. R/W pin is connected to ground to make LCD1 write always. A 1-kilo-ohm preset (VR1) is connected to VO pin of LCD1 to vary its contrast. Pin 15 of LCD1 is connected to +5V through 220-ohm resistor R10 and pin 16 to ground to turn on the back light of LCD1.

Port-B pins PB0 through PB3 (pins 1 through 4) of IC1 are connected to input pins of IC2. DC motors are connected to output pins of IC2.

Port-B pins PB4 through PB7 (pins 5 through 8) of IC1 drive the four LEDs through a 220-ohm current-limiting resistor each. AVCC (pin 30) and AREF (pin 32) of IC1 are given 5V to supply voltage and reference voltage for the internal ADC. A 10-kilo-ohm resistor (R5) is connected between Vcc and reset pin 9 (RST) of IC1. To make reset pin active-low, pushbutton switch S1 is connected between pin 9 and ground to apply active-low reset signal to IC1. Fuse bits are set such that IC1 works with the internal RC oscillator frequency of 1MHz. So you need not connect any external crystal at pins XTAL1 (pin 12) and XTAL2 (pin 13) of IC1.

Flex sensors, LCD1, ATmega32 and L293D all are given 5V, while L293D works off 12V supply.

Software

Complete working of the circuit is controlled by the software program (flex.c), which is already loaded into the internal flash memory of ATmega32 microcontroller. The program is written in 'C' language using AVR Studio software for the AVR microcontroller, and compiled and simulated using AVR simulator available with AVR Studio.

The program scans all the four ADC channels of IC1 and converts analogue input voltages from flex sensors FS1 through FS4 into digital equivalent hex values, and further into decimal values. The decimal value is around 700 when the flex sensor is straight and 850-900 when the flex sensor is completely bent. So the threshold limit is set to 820.

If none of the sensor values is more than